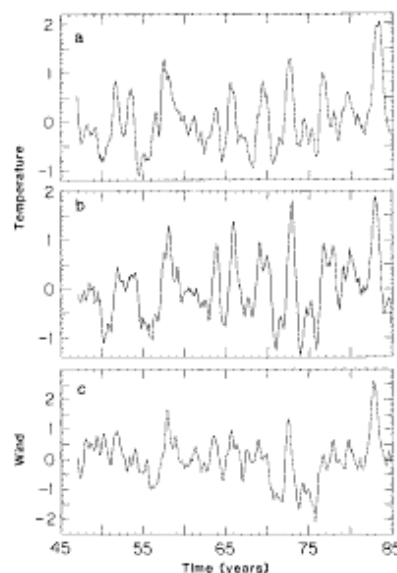


Fig. 1. (a and b) Time series for 1947 to 1985, of average sea-surface temperature anomaly ($^{\circ}\text{C}$, annual cycle removed) for (a) the eastern equatorial Pacific (10°N to 10°S , 100°W to 80°W) and (b) the central equatorial Pacific (170°W to 120°W , 5°S to 5°N). (c) Time series of eastward surface wind anomaly (m sec^{-1}) in the central equatorial Pacific. Winds are combined averages over regions 150°E to 140°W , 0° to 10°N and 160°E to 160°W , 0° to 10°S . All series are of 5-month running means. The overall similarity of the fields is suggestive of large-scale controlling processes, and thus indicates that a low-order model can capture much of the essential dynamics. [Data courtesy of T. Barnett]



wind across the ocean, because of the convective tendency for air to rise (sink) over warm (cool) water. Thus we write

$$\frac{du}{dt} = B(T_e - T_w)/2\Delta x - C(u - u^*) \quad (1)$$

where B and C are constants. The terms $B(T_e - T_w)/2\Delta x + Cu^*$ represent wind-produced stress, $-Cu$ represents mechanical damping, and a negative value for the constant u^* would represent the effect of the mean tropical surface easterlies. (Variations in pressure have been neglected; they do not qualitatively affect the model's behavior.) The temperature field is advected by the current. Assuming a deep ocean of constant temperature $\bar{T}$, the simplest finite difference approximation to the temperature equation of fluid flow is

$$\frac{dT_w}{dt} = \frac{u}{2\Delta x}(\bar{T} - T_e) - A(T_w - T^*) \quad (2)$$

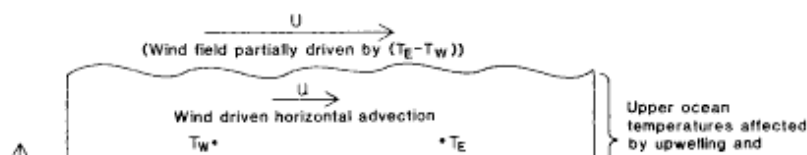
$$\frac{dT_e}{dt} = \frac{u}{2\Delta x}(T_w - \bar{T}) - A(T_e - T^*) \quad (3)$$

The first term on each right-hand side represents horizontal advection and upwelling and the second forcing and thermal damp-

ing (A and T^* are constants). T^* is the temperature to which the ocean would relax in the absence of motion, and therefore represents radiative processes and heat exchange with the atmosphere. Equations 1 to 3 constitute the basic model. Obviously some physical processes have been neglected. In particular, small-scale structures, certain types of wave propagation, and the role of absolute temperature (in latent heat release) have not been represented. Kelvin waves may be important in transmitting the influence of a central equatorial wind stress to the eastern ocean (8-10). They can be considered a pressure wave whose purpose is to maintain a divergence-free flow. In the model presented here, upwelling is directly

related to a horizontal current through the continuity relation, and in this limited sense the need for a Kelvin wave is obviated. Also, the presence of an equatorial wave-guide is implicitly recognized through the restriction of motion to a zonal plane at the equator. Furthermore, the effects of, but no feedback onto, a Hadley circulation (a primarily meridional convective cell in the tropics that produces the mean surface easterlies) are present through the term u^* . Thus, much of the essential physics of large-scale air-sea interactions in the tropics has been preserved.

Without loss of generality, we set $\bar{T} = 0$, so measuring all temperatures with respect to $\bar{T}$. For $u^* = 0$ it is possible to obtain some qualitative insight to the model's behavior, because some mathematical analysis is possible. Steady solutions are found by setting the left-hand sides of Eqs. 1 to 3 equal to zero and solving the resulting equations. For $B' < AC/T^*$ (where $B' = B/2\Delta x^2$), only one real solution exists. Because the influence of the ocean temperature on the wind is so small, the state is stable and has $u = 0$ and $T_e = T_w = T^*$. If we let B increase (keeping other parameters constant), the system will bifurcate when $B' = AC/T^*$ (it is actually a pitchfork bifurcation). The existing solution becomes unstable and two new, real, stable solutions appear. All the solutions become unstable for $B' > (4A + C)C^2/T^*(C - 2A)$. (The system undergoes a subcritical Hopf bifurcation at this point.) However, the system is bounded (meaning that T_w , T_e , and u cannot exceed certain values). Three-dimensional bounded dynamical systems with quadratic nonlinearity can display chaotic, aperiodic behavior (15) [whereas systems with two dependent variables (16) are restricted to stationary or limit-cycle behavior unless time-dependent forcing is added]. Indeed, with $u^* \neq 0$, Eqs. 1 to 3 have the same structure as the Lorenz equations. The physical system resembles that of ordinary convection in that the velocity of circulation is maintained by a horizontal temperature gradient: this sets up a direct circulation in the



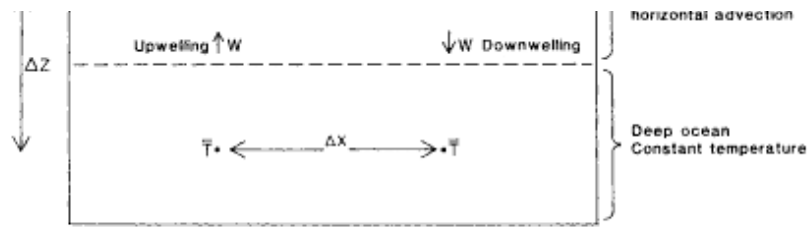


Fig. 2. Schema of model. Below the "thermocline" the temperature is constant, T^* . The wind, driven in part by the temperature gradient $(T_e - T_w)$, generates a current which, by mass continuity, generates upwelling or downwelling as well as horizontal advection. Now if $\partial T/\partial t = -(wT)_x - (wT)_x$ and normal velocities are zero at domain boundaries, then $\partial T_w/\partial t = \{0 - w(T_w + T_e)/2\}/\Delta x + \{w(T^* + T_w)/2 - 0\}/\Delta z = w(T^* - T_e)/2\Delta x$, since $w\Delta z = w\Delta x$ by mass conservation. Addition of damping terms gives Eq. 2. Similarly for T_e .

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atmosphere, which forces the ocean. The unusual aspect here is that the oceanic part of the system is able to overturn with warmer water above the cool deep ocean—a conventionally stable configuration because work must be done on the system. Overturning and instability can occur here because the dominant driving of the ocean is mechanical, namely wind stress.

Thus for constant T^* , C , and A , all stationary solutions become unstable when B , the influence of ocean temperatures on the wind, is sufficiently large. The system can no longer stably reside anywhere and oscillates

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